HistogramObserver#

https://pytorch.org/docs/stable/generated/torch.ao.quantization.observer.His

Description:

The module records the running histogram of tensor values along with min/max values. calculate_qparams will calculate scale and zero_point. bins (int) – Number of bins to use for the histogram dtype (dtype) – dtype argument to the quantize node needed to implement the reference model spec qscheme – Quantization scheme to be used reduce_range – Reduces the range of the quantized data type by 1 bit

Function Signature

```
class
torch.ao.quantization.observer.HistogramObserver(bins=2048,
dtype=torch.quint8, qscheme=torch.per_tensor_affine,
reduce_range=False, quant_min=None, quant_max=None,
factory_kwargs=None, eps=1.1920928955078125e-07,
is_dynamic=False, **kwargs)[source]#

Parameters

Create the histogram of the incoming inputs.

Search the distribution in the histogram for optimal min/max
values.

Compute the scale and zero point the same way as in the
```

Detailed Documentation

Documentation

The module records the running histogram of tensor values along with min/max values. calculate_qparams will calculate scale and zero_point.

bins (int) – Number of bins to use for the histogram

dtype (dtype) – dtype argument to the quantize node needed to implement the reference model spec

qscheme – Quantization scheme to be used

reduce_range – Reduces the range of the quantized data type by 1 bit

eps (Tensor) – Epsilon value for float32, Defaults to torch.finfo(torch.float32).eps.

The scale and zero point are computed as follows:

The histogram is computed continuously, and the ranges per bin change with every new tensor observed.

The search for the min/max values ensures the minimization of the quantization error with respect to the floating point model.